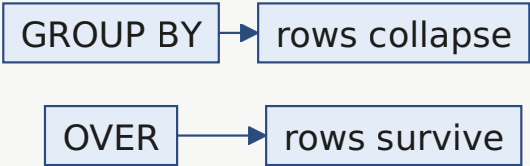


1 Collapse or keep



If the answer needs the original row back, it is a window. If the answer is one number per group, it is `GROUP BY`, and reaching for a window is over-engineering.

Crux. `GROUP BY` answers how much per group and the rows disappear; a window answers where this row stands and every row survives.

2 OVER has two dials

```
rank() OVER (PARTITION BY segment ORDER BY revenue DESC)
```

These are not decoration on a function name. They are the definition of the question being asked.

Crux. `PARTITION BY` says who counts as neighbours and the window's own `ORDER BY` says in what order, and changing either changes the answer.

3 Three functions, one tie

Function	On a tie	After it	Repeatable
ROW_NUMBER	Breaks it arbitrarily	1 2 3	No
RANK	Shares a position	1 1 3	Yes
DENSE_RANK	Shares a position	1 1 2	Yes

`RANK` answers how many are ahead of me. `DENSE_RANK` answers how many levels are ahead of me.

Crux. The difference only shows up on a tie, and a tie at the boundary changes how many rows your report ships.

4 Top-N inside a group

```
WITH r AS (  
  SELECT segment, customer_id, revenue,  
         rank() OVER (PARTITION BY segment  
                      ORDER BY revenue DESC) AS pos  
  FROM q2  
)  
SELECT * FROM r WHERE pos <= 50;
```

`LIMIT 50` takes fifty rows from the whole result, never fifty from each group.

Crux. A window cannot be filtered in `WHERE`, so compute it inside and filter outside.

5 LAG, and what NULL means

```
lag(spend, 1) OVER (PARTITION BY customer_id  
                   ORDER BY month)
```

Falling two months running is two LAGs and a comparison. A member with one month of data returns `NULL`, fails the comparison and drops out, which is honest: you cannot tell.

Crux. `LAG` reads the previous row inside the partition, and the first row of every partition returns `NULL` because there is nothing behind it.

6 Running totals and determinism

```
sum(revenue) OVER (ORDER BY week_start)
```

If two rows share `week_start`, their relative order is undefined and the cumulative column can differ between runs. Add a tiebreaker. A number that changes between runs is worse than one that is wrong, because nobody can reproduce the argument about it.

Crux. A running total is only reproducible when the window's order cannot tie.